COM3021

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

COMPUTER SCIENCE

Examination, May 2022

Data Science at Scale

Module Leader: Dr. Hugo Barbosa

Duration: 2 HOURS

- All questions are compulsory.
- The word estimates are provided as a guidance for the expected length of an effective and concise answer to the questions.
- This exam is worth 70% of the overall module grade.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

Question 1

(a) In the context of large-scale data systems, provide a definition of **Reliability**, taking into account aspects such as user experience, system performance and security. *(a satisfactory answer will have between 20 and 50 words)*

(10 marks)

(b) In a large-scale distributed system, it is certain that components will fail. In this context, compare **software** and **hardware** faults, and their impact to the resilience of the system. *(a satisfactory answer will have between 20 and 50 words)*

(10 marks)

(Total 20 marks)

Question 2

- Explain **one advantage** and **one disadvantage** of a document store in comparison to a traditional relational database. *(a satisfactory answer will have between 20 and 50 words)*

(10 marks)

- Explain **two benefits** and **two drawbacks** of **replication** in the context of distributed architectures. *(a satisfactory answer will have between 20 and 50 words)*

(10 marks)

(Total 20 marks)

Question 3

You are designing a distributed data system for a social media platform. Which data distribution architecture would deliver a more scalable, reliable and resilient system? Justify your choice describing the advantages of your approach against other possible strategies. *(a satisfactory answer will have between 50 and 80 words)*

(10 marks)

(Total 10 marks)

Question 4

You were tasked to design the architecture of a new data system that receives COVID test data submitted every day by approximately 2,000 hospitals from all over the country.

(a) Your system will generate daily outputs from the collected results and produce aggregated statistics from it. What choices would you make regarding the design of the data system for this solution, taking into account aspects such as **scalability**, **reliability** and **data processing**? Explain and justify your design choices. *(a satisfactory answer will have between 60 and 90 words)*

(15 marks)

(b) One component of your system will provide real-time data as the test data arrive. This data will be consumed by a dashboard application showing live statistics and predictions to key decision-makers. How would you implement the server-side of this system and how would this component interact with other components (e.g., the data storage)? *(a satisfactory answer will have between 30 and 60 words)*

(10 marks)

(c) Later on it is decided that this system will provide an integration with social media platforms, allowing users to post and share the daily statistics and predictions. Another member of the team suggests using MPI to implement the social media integration through message-passing. How would you evaluate this suggestion? Justify your answer. *(a satisfactory answer will have between 10 and 30 words)*

(5 marks)

(Total 30 marks)

Question 5

(a) In the context of distributed systems, explain the CAP theorem and the trade-off it entails. *(a satisfactory answer will have between 30 and 60 words)*

(6 marks)

(b) Describe *vertical* and *horizontal* scaling and three key differences between them, taking into account scalability, resilience and maintainability aspects. *(a satisfactory answer will have between 60 and 90 words)*

(14 marks)

(Total 20 marks)